

Shambhav Kumar

📞 +91-7543912012 | ✉️ shambhavkumar2002@gmail.com
https://github.com/Itzshambhav | https://www.linkedin.com/in/shambhav-kumar-2aaa30212/

Professional Summary

Computer Science undergraduate passionate about Backend Development, AI-powered applications, and scalable software systems. Skilled in Python, FastAPI, REST APIs, Docker, and NLP-based applications.

Projects

AI-Teacher Chatbot | Python, FastAPI, NLP, LangChain | [GitHub Link](#)

- Developed a multilingual AI chatbot using **LangChain** and **LLaMA 3.1** for intelligent question-answering.
- Built an interactive web interface with **Streamlit** to enable real-time user conversations.
- Integrated conversational memory to maintain chat history and provide context-aware responses.
- Designed custom prompt engineering techniques for structured and educational AI-generated answers.
- Implemented Hugging Face model integration and optimized response generation for better user experience.

Video Cryptography | Java, Spring Boot, JUnit, Mockito, GitHub Actions, Docker | [GitHub Link](#)

- Built full-stack web application using Java and Spring Boot for secure video encryption and decryption.
- Designed REST APIs for secure file transfer and protected data communication.
- Implemented unit and integration testing using JUnit and Mockito to improve application reliability.
- Built CI/CD workflows using GitHub Actions for automated testing and deployment.
- Containerized the application using Docker for scalable and consistent deployment environments.

Image Denoising System (DnCNN) | Python, FastAPI | [GitHub Link](#)

- Implemented a deep learning-based image denoising model using DnCNN architecture.
- Built REST APIs using FastAPI for image upload and real-time processing.
- Performed image preprocessing and model optimization for improved output quality.
- Focused on enhancing model reliability and inference performance.
- Explored practical applications of deep learning in computer vision tasks.

Skills

Programming Languages: C++, Python, SQL

Frontend : HTML, CSS, JavaScript, React.js

Backend: FastAPI, Django, SpringBoot, Streamlit, Git

AI / ML: Machine Learning, Deep Learning, LangChain, FAISS, NLP

Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, System Design, DBMS

Databases: MySQL, MongoDB

Cloud & DevOps: AWS (Basics), CI/CD, Docker, Deployment

Architecture: Microservices, Scalable Systems

Education

Graphic Era Hill University

B.Tech in Computer Science and Engineering, CGPA : 7.38/10

Dehradun, India

2021 – 2026

Sri Chaitanya

12th (C.B.S.E), Percentage : 79%

Vishakhapatnam, India

2019 – 2020

P.P.J. Saraswati Vihar

10th (C.B.S.E), Percentage : 86.5 %

Nainital, Uttarakhand

2017 – 2018

Certifications & Achievements

- Participated in **Adobe India Hackathon 2025** – Round 1 Online MCQ Assessment & Coding Challenge.
- Completed **Web Development Virtual Internship at CodSoft (2025)**.
- Participated in **GDG on Campus Solution Challenge 2025** powered by Google Developer Groups and Hack2skill.
- Strong foundation in **Data Structures & Algorithms and problem-solving**.
- Earned **NSS 'B' and 'C' Certificates** for leadership, volunteering, and social awareness activities.
- Volunteer at **38th National Games Uttarakhand 2025**, contributing to event coordination and teamwork activities.